FACILITATION GUIDE

LESSON TITLE: Introduction to Information Technology Written Exam

LESSON DESIGNATOR: DS.24.01

TOTAL LESSON TIME: 2 Hours

LEARNING OBJECTIVES:

ELOs:

1. Without the aid of reference, identify in writing the fundamentals of networking models, in accordance with ISBN: 978-1260122381. (0671-IMPL-1002ce).
2. Without the aid of reference, identify in writing the fundamentals of internet protocol environments, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002cf).
3. Without the aid of of reference, identify in writing the fundamentals of network enclaves, in accordance with ISBN-13: 978-1260454031. (0671-IMPL-1002cg).
4. Without the aid of reference, identify in writing the fundamentals of wiring standards, in accordance with ISBN:978-1-118-80732-3. (0671-IMPL-1002ch).
5. With the aid of reference and given the proper equipment and software, backup active directory data, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002ci).
6. Without the aid of reference, identify in writing the functions of a service desk, in accordance with ISBN-13: 978-0-324-78506-7. (0671-IMPL-1002cj).
7. Without the aid of reference, identify in writing the fundamentals of programs of records, in accordance with MCRP 3-30B.2. (0671-IMPL-1002ck).
8. Without the aid of reference, identify in writing the foundational cybersecurity requirements, in accordance with ISBN: 1-939136-05-3. (0671-IMPL-1002cl).
9. Without the aid of reference, identify in writing the characteristics of cabling standards, in accordance with ISBN: 978-1-118-80732-3. (0671-IMPL-1002cm).
10. Without the aid of reference, identify in writing the characteristics of copper cabling, in accordance with ISBN: 978-1-118-80732-3. (0671-IMPL-1002cn).
11. Without the aid of reference, identify in writing the types of copper cables, in accordance with ISBN: 978-1-118-80732-3. (0671-IMPL-1002co).
12. Without the aid of reference, identify in writing the characteristics of fiber cabling, in accordance with ISBN: 978-1-118-80732-3. (0671-IMPL-1002cp).
13. Without the aid of reference, identify in writing the types of fiber cables, in accordance with ISBN: 978-1-118-80732-3. (0671-IMPL-1002cq).
14. Without the aid of reference, identify in writing the characteristics of twisted pair cabling, in accordance with ISBN: 978-1-118-80732-3. (0671-IMPL-1002cr).

15. Without the aid of reference, identify in writing the types of twisted pair cabling, in accordance with ISBN: 978-1-118-80732-3. (0671-IMPL-1002cs).

16. Without the aid of reference, identify in writing the characteristics of wireless media, in accordance with ISBN:978-1-118-80732-3. (0671-IMPL-1002ct).

17. Without the aid of reference, identify in writing the characteristics of internet protocol addresses, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002cu).

18. Without the aid of reference, identify in writing the components of an internet protocol address, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002cv).

19. Without the aid of reference, identify in writing an internet protocol network, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002cx).

20. Without the aid of reference, identify in writing the characteristics of media access control addresses, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002cy).

21. Without the aid of reference, identify in writing the characteristics of internet protocol networks, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002cz).

22. Without the aid of reference, identify in writing the subnet of a network, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002da).

23. Without the aid of reference, identify in writing Variable Length Subnet Mask, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002db).

24. Without the aid of reference, identify in writing the characteristics of subnet masks, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002dc).

25. Without the aid of reference, identify in writing the characteristics of Classless Inter Domain Routing notation, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002dd).

26. Without the aid of reference, identify in writing a subnet mask, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002de).

27. Without the aid of reference, identify in writing the characteristics of binary notation, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002df).

28. Without the aid of reference, identify in writing the characteristics of decimal notation, in accordance with ISBN:978-1-119-35914-2. (0671-IMPL-1002dg).

29. Without the aid of reference, convert in writing binary addresses into decimal notation, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002dh).

30. Without the aid of reference, identify in writing the implementation of the security configuration for the programs of record, in accordance with MCRP 3-30B.2. (0671-IMPL-1002di).

31. Without the aid of reference, identify in writing the maintenance procedures of the security configuration for the programs of record, in accordance with MCRP 3-30B.2. (0671-IMPL-1002dj).

32. Without the aid of reference, convert in writing decimal notation to binary notation, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002dk).

33. Without the aid of reference, identify in writing the characteristics of the osi model, in accordance with ISBN:

978-1260122381. (0671-IMPL-1002dl).

34. Without the aid of reference, identify in writing the components of Marine Corps common hardware suite, in accordance with MCRP 3-30B.2. (0671-IMPL-1002dm).

35. Without the aid of reference, identify in writing the purpose of maintenance procedures, in accordance with MCRP.3-30B.2. (0671-IMPL-1002dn).

36. Without the aid of reference, identify in writing the characteristics of the networking models, in accordance with ISBN: 978-1260122381. (0671-IMPL-1002do).

37. Without the aid of reference, define in writing the physical layer of the osi model, in accordance with ISBN: 978-1260122381. (0671-IMPL-1002dp).

38. Without the aid of reference, define in writing the data link layer of the osi model, in accordance with ISBN: 978-1260122381. (0671-IMPL-1002dq).

39. Without the aid of reference, define in writing the network layer of the osi model, in accordance with ISBN: 978-1260122381. (0671-IMPL-1002dr).

40. Without the aid of reference, define in writing the transport layer of the osi model, in accordance with ISBN: 978-1260122381. (0671-IMPL-1002ds).

41. Without the aid of reference, define in writing the session layer of the osi model, in accordance with ISBN: 978-1260122381. (0671-IMPL-1002dt).

42. Without the aid of reference, define in writing the presentation layer of the osi model, in accordance with ISBN: 978-1260122381. (0671-IMPL-1002du).

43. Without the aid of reference, define in writing the application layer of the osi model, in accordance with ISBN: 978-1260122381. (0671-IMPL-1002dv).

44. Without the aid of reference, identify in writing the purpose of network enclaves, in accordance with ISBN-13: 978-1260454031. (0671-IMPL-1002dw).

45. Without the aid of reference, identify in writing the types of network enclaves, in accordance with ISBN-13: 978-1260454031. (0671-IMPL-1002dx).

46. Without the aid of reference, identify in writing the characteristics of local area networks, in accordance with ISBN-13: 978-1260454031. (0671-IMPL-1002dz).

47. Without the aid of reference, identify in writing the characteristics of wide area networks, in accordance with ISBN-13: 978-1260454031. (0671-IMPL-1002ea).

48. Without the aid of reference, identify in writing the characteristics of virtual local area networks, in accordance with ISBN-13: 978-1260454031. (0671-IMPL-1002eb).

49. Without the aid of reference, identify in writing the depictions of intermediary device symbols, in accordance with MCRP 3-30B.2. (0671-IMPL-1002ec).

50. Without the aid of reference, identify in writing the depictions of end-device symbols, in accordance with MCRP 3-30B.2. (0671-IMPL-1002ed).

51. Without the aid of reference, identify in writing the depictions of network media symbols, in accordance with MCRP 3-30B.2. (0671-IMPL-1002ee).

52. Without the aid of reference, identify in writing the characteristics of a physical network diagram, in accordance with MCRP 3-30B.2. (0671-IMPL-1002ef).

53. Without the aid of reference, identify in writing the characteristics of a logical network diagram, in accordance with MCRP 3-30B.2. (0671-IMPL-1002eg).

54. Without the aid of reference, identify in writing the types of network diagram, in accordance with MCRP 3-30B.2.(0671-IMPL-1002eh).

55. Without the aid of reference, identify in writing the characteristics of service desks, in accordance with ISBN-13:978-0-324-78506-7. (0671-IMPL-1002ei).

56. Without the aid of reference, identify in writing the purpose of service desks, in accordance with ISBN-13: 978-0-324-78506-7. (0671-IMPL-1002ej).

57. Without the aid of reference, identify in writing the best practices of service desks, in accordance with ISBN-13: 978-0-324-78506-7. (0671-IMPL-1002ek).

REFERENCES:

Reference Title	Publication ID
A GUIDE TO HELP DESK CONCEPTS, THIRD EDITION	ISBN-13: 978-0-324-78506-7
CABLING: THE COMPLETE GUIDE TO COPPER AND FIBER-OPTIC NETWORKING	ISBN: 978-1-118-80732-3
CompTIA A+ Certification All-in-One Exam Guide, Eleventh Edition (Exams 220-1101 & 220-1102)	ISBN-13: 978-1260454031
COMPTIA NET+ CERTIFICATION ALL-IN-ONE EXAM GUIDE, SEVENTH EDITION (EXAM N10-007)	ISBN: 978-1260122381
COMPTIA SECURITY+: GET CERTIFIED GET AHEAD SY0-501 STUDY GUIDE	ISBN: 1-939136-05-3
Cybersecurity	DoDI 8500.01
DISA Security Technical Implementation Guides	DISA STIGS
MAGTF Communications System	MCRP 3-30B.2
Marine Corps Cybersecurity	MCO 5239.2B
Marine Corps Enterprise Cybersecurity Manual Library	USMC ECSM Library
MASTERING MICROSOFT EXCHANGE SERVER 2016	ISBN: 978-1-119-23205-6
MCSA WINDOWS 2016 COMPLETE STUDY GUIDE	ISBN: 978-1-119-35914-2
Risk Management Framework (RMF) for DoD Information Technology (IT)	DoDI 8510.01

PERSONNEL REQUIRED: 3 facilitators; 1 Course Chief, 1 Curriculum Developer

FACILITIES REQUIRED: Bldg 1763 Room 250

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MATERIALS/EQUIPMENT:

Presentation Software PS	Instructors Intelligence Workstation (I) IW
Student Intelligence Workstation (S) IW	Admin Time AD
Check-in CHKIN	Checkout CKOUT
Computer Workstation w/ Software & Network Access CWS(N)	Dry Erase Board DB
End of Course Critique EOCC	Examination EXAM
Graduation Certificate Grad Cert	Learning Management System LMS
Lesson Plan LP	Laptop Computer LTC
Performance Evaluation Checklist PECL	Performance Exam PERF EXAM
PowerPoint Presentation PPT	Reference Materials (Digital and/or Paper Based) REF
Smart Board SB	Student Handout SH
Tablet TAB	Training Schedule TS

LEARNING EXPERIENCE ACTIVITIES:

Method	**Code**
Academic Orientation	ACOR
Administrative	ADMIN
Coaching	CCH
Course Introduction	CI
Check-in	CKIN
Check-Out	CKOUT
Demonstration	D
End of Course Critique	EOCC
Guided Discussion	GD
Graduation	GRAD
Gallery Walk	GW
Homework	HW
Informal Lecture	L
Informal Lecture/Guided Discussion	IL/GD
Informal Lecture/Gamification	IL/GM
Informal Lecture/Jigsaw	IL/JSW
Informal Lecture/Questioning	IL/Q
Independent Study Time	IST
Leadership Guided Discussion	LDGD
Practical Application	PA
Quiz	Q
Socratic Questioning	SOQ
Teach Back	TB
Think-Pair-Share	TPS
Performance Exam	X(P)
Exam Review	X(RE)
Written Examination	X(W)

Learner Role(s): **Data Systems Administrator**

The Data Systems Administrator is a critical role in the 21st century digital landscape, responsible for designing, implementing, and managing data systems that support organizational operations. This role

requires a deep understanding of the fundamentals of information technology, as well as the ability to apply 21st century learning concepts such as critical thinking, problem-solving, and collaboration.

Key Responsibilities:

- Design and implement data systems that meet organizational needs and requirements.
- Manage and maintain data systems, including hardware, software, and networks.
- Troubleshoot and resolve data system issues, using critical thinking and problem-solving skills.
- Collaborate with stakeholders to identify and prioritize data system needs and requirements.
- Develop and implement data backup and recovery plans, ensuring data integrity and security.
- Stay up to date with emerging trends and technologies in data systems administration, applying 21st century learning concepts such as innovation, creativity, and adaptability.

21st Century Learning Concepts:

- Critical thinking: analyzing complex data system issues and developing effective solutions.
- Problem-solving: troubleshooting and resolving data system problems, using creative and innovative approaches
- Collaboration: working with stakeholders to identify and prioritize data system needs and requirements
- Communication: effectively communicating data system information and requirements to stakeholders
- Creativity: designing and implementing innovative data systems solutions that meet organizational needs
- Innovation: staying up-to-date with emerging trends and technologies in data systems administration, and applying them to improve organizational operations

Learning Objectives:

- Understand the fundamentals of information technology, including data systems, networks, and cybersecurity.
- Analyze complex data system issues and develop effective solutions, using critical thinking and problem-solving skills.
- Design and implement data systems that meet organizational needs and requirements, using collaboration and communication skills.
- Troubleshoot and resolve data system issues, using creative and innovative approaches.
- Develop and implement data backup and recovery plans, ensuring data integrity and security.
- Stay up-to-date with emerging trends and technologies in data systems administration, applying 21st century learning concepts such as innovation, creativity, and adaptability.

Performance Tasks:

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- Design and implement a data system for a fictional organization, using critical thinking and problem-solving skills.
- Troubleshoot and resolve a complex data system issue, using creative and innovative approaches.
- Develop and implement a data backup and recovery plan, ensuring data integrity and security.
- Collaborate with stakeholders to identify and prioritize data system needs and requirements, using communication and collaboration skills.
- Create a presentation or report on emerging trends and technologies in data systems administration, applying 21st century learning concepts such as innovation, creativity, and adaptability.

Resources:

- Online learning platforms and resources, such as video tutorials and interactive simulations
- Textbooks and e-books on data systems administration and information technology
- Industry publications and research articles on emerging trends and technologies in data systems administration
- Collaboration tools, such as discussion boards and group projects, to facilitate communication and teamwork.

Facilitator Role: The Data Systems Administration Instructor plays a crucial role in delivering instruction and guiding learners in developing knowledge and skills related to data systems administration, including hardware, software, networks, and cybersecurity. To achieve this, the instructor employs a range of instructional methods, including lectures, discussions, and hands-on activities, to engage learners and promote understanding. By facilitating group discussions, providing individualized support, and using technology such as virtual labs and online resources, the instructor creates a positive and inclusive learning environment that encourages active learning and participation. The instructor's key responsibilities include delivering instruction, guiding learners in completing performance tasks and projects, and evaluating learner progress, all while ensuring learners understand the fundamentals of information technology and can apply their knowledge to troubleshoot and resolve data system issues. Ultimately, the instructor's effectiveness is measured by learner engagement, understanding, and application of data systems administration concepts, as well as their ability to adapt instruction to meet learner needs and promote understanding.

LEARNING EXPERIENCE REFLECTION: Describe your reflection method. What feedback will you provide?

As I reflect on my learning experience, I will provide feedback on the following aspects:

- The effectiveness of the instructional methods and materials used in the course.
- The clarity and relevance of the learning objectives and outcomes

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- The level of support and guidance provided by the instructor and peers.
- The opportunities for practice and application of the concepts and skills learned.
- The overall quality of the learning experience and its impact on my knowledge and skills in Data Systems Administration

I will provide constructive feedback that is specific, timely, and respectful, with the goal of improving the learning experience for future learners. My feedback will include suggestions for improvements, such as additional resources or support, and recommendations for how to build on the strengths of the course. By providing thoughtful and reflective feedback, I aim to contribute to the ongoing improvement and refinement of the Data Systems Administration course, and to help ensure that future learners have a high-quality and effective learning experience.

ASSESSMENT: How do learners demonstrate their understanding?

- Quizzes and exams to assess understanding of the fundamentals of information technology.
- Performance tasks and projects to assess application of critical thinking, problem-solving, and collaboration skills.
- Peer review and feedback to assess communication and collaboration skills.

SAFETY:

☐ Review/Reassess the learning environment/location for changes that affect the original Risk Assessment.

APPROVING SIGNATURE _____DATE _____